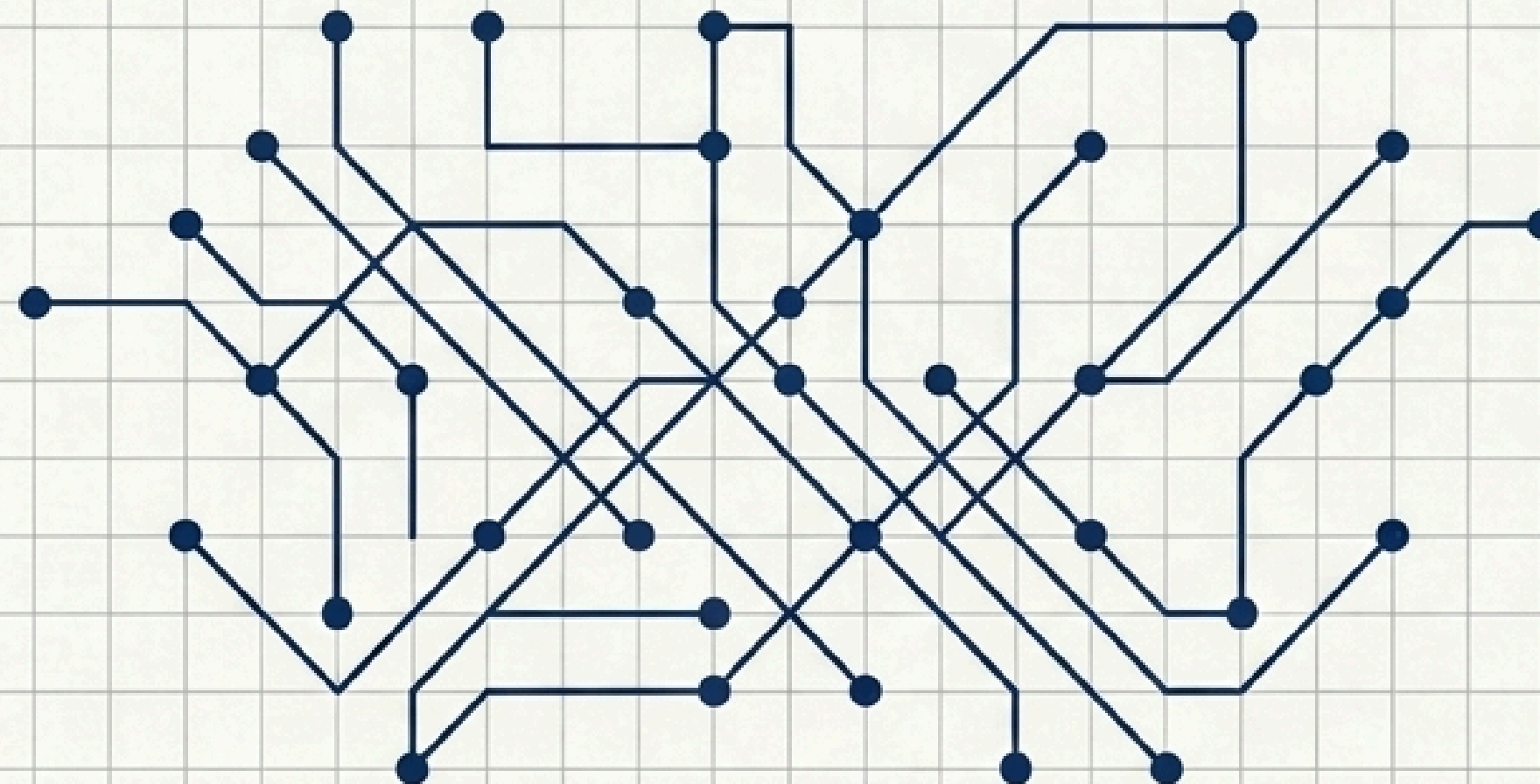




Educational Volunteer Coordination Platform

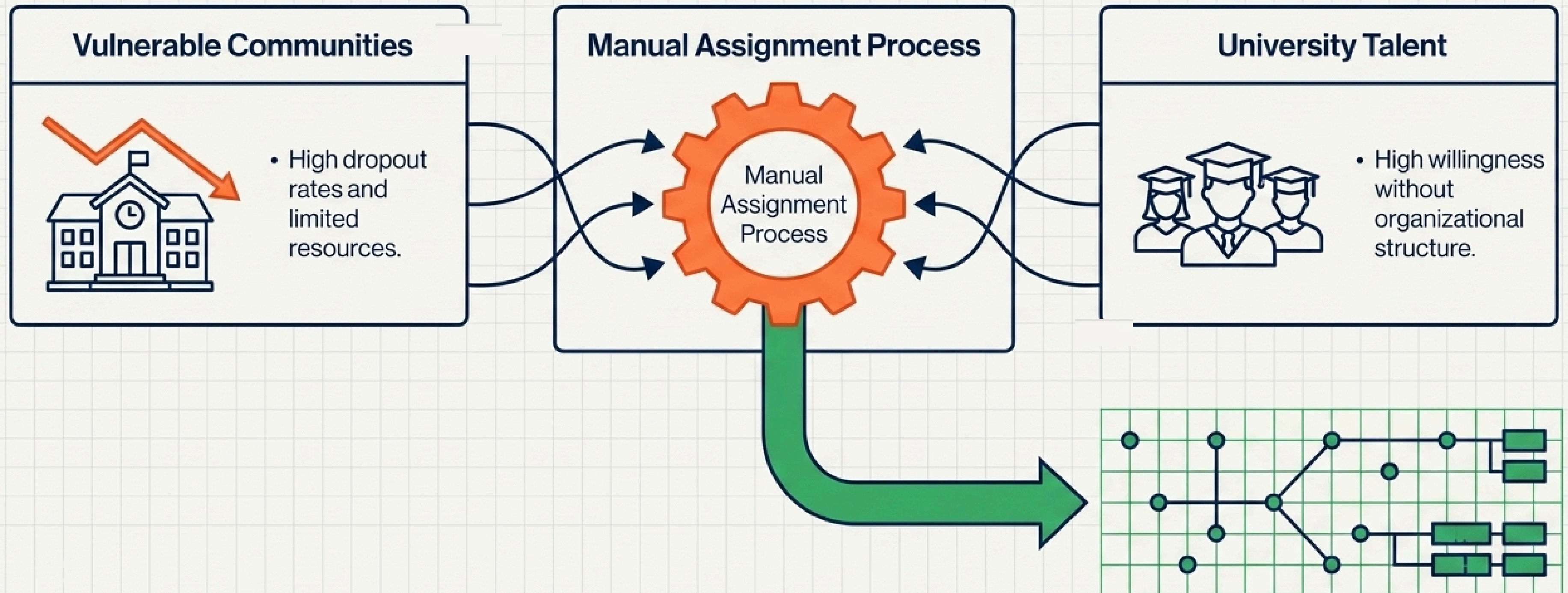
Authors: Jose Luis Leudo Mosquera, David Santiago Rojas Hernández, Adrián Mauricio Hernández Ortiz, Johan Sebastian Candia Marín

University: Universidad Distrital Francisco José de Caldas

Advisor: Professor Carlos Andrés Sierra, M.Sc.

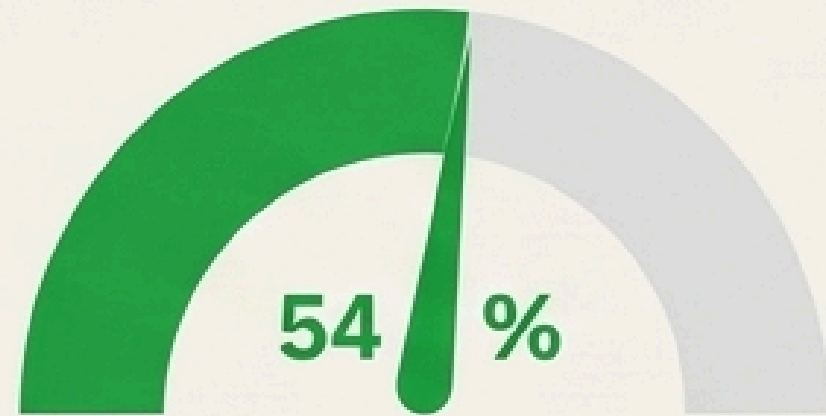
The Social Context: The Intersection of Need and Talent

Transforming manual assignment into a traceable digital ecosystem.



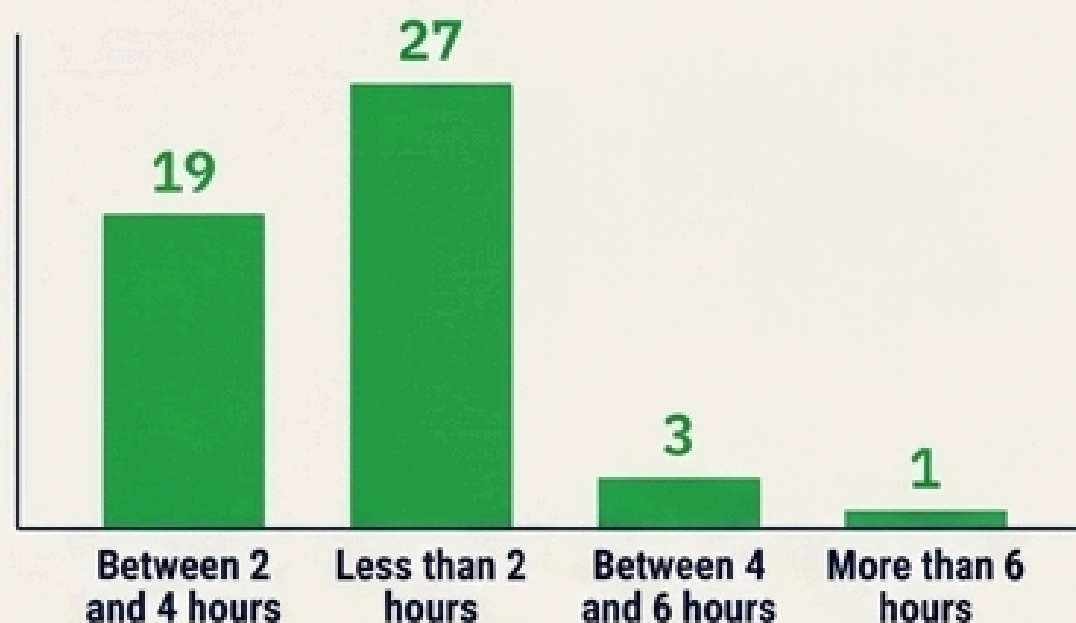
Listening to the User: Baseline Survey Findings (N=50)

Empirical data directly dictates the architectural constraints of the system.

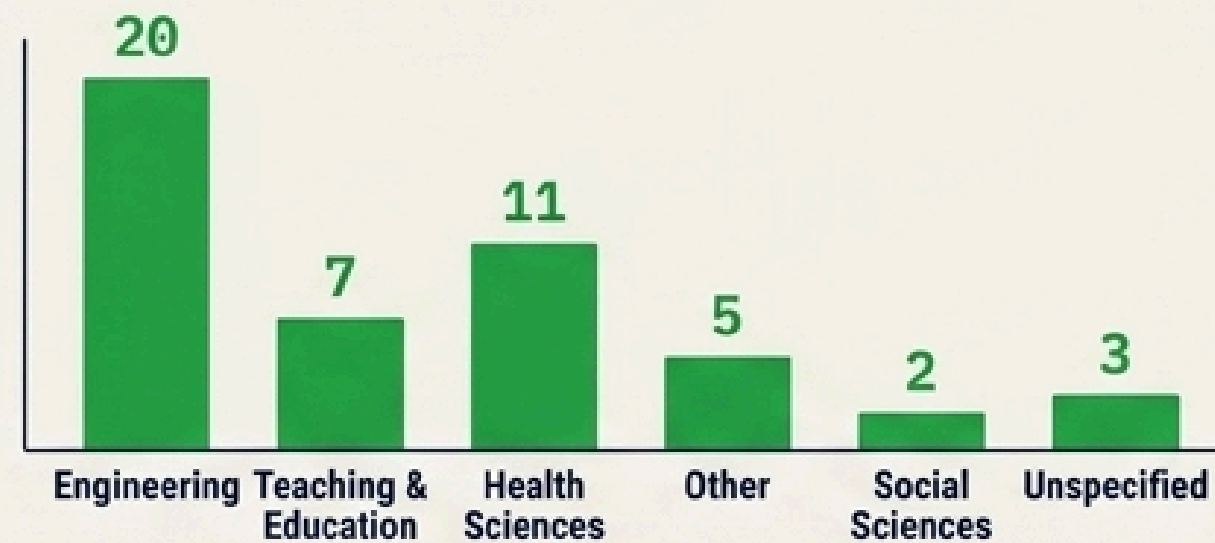


54% possess < 2 hours/week

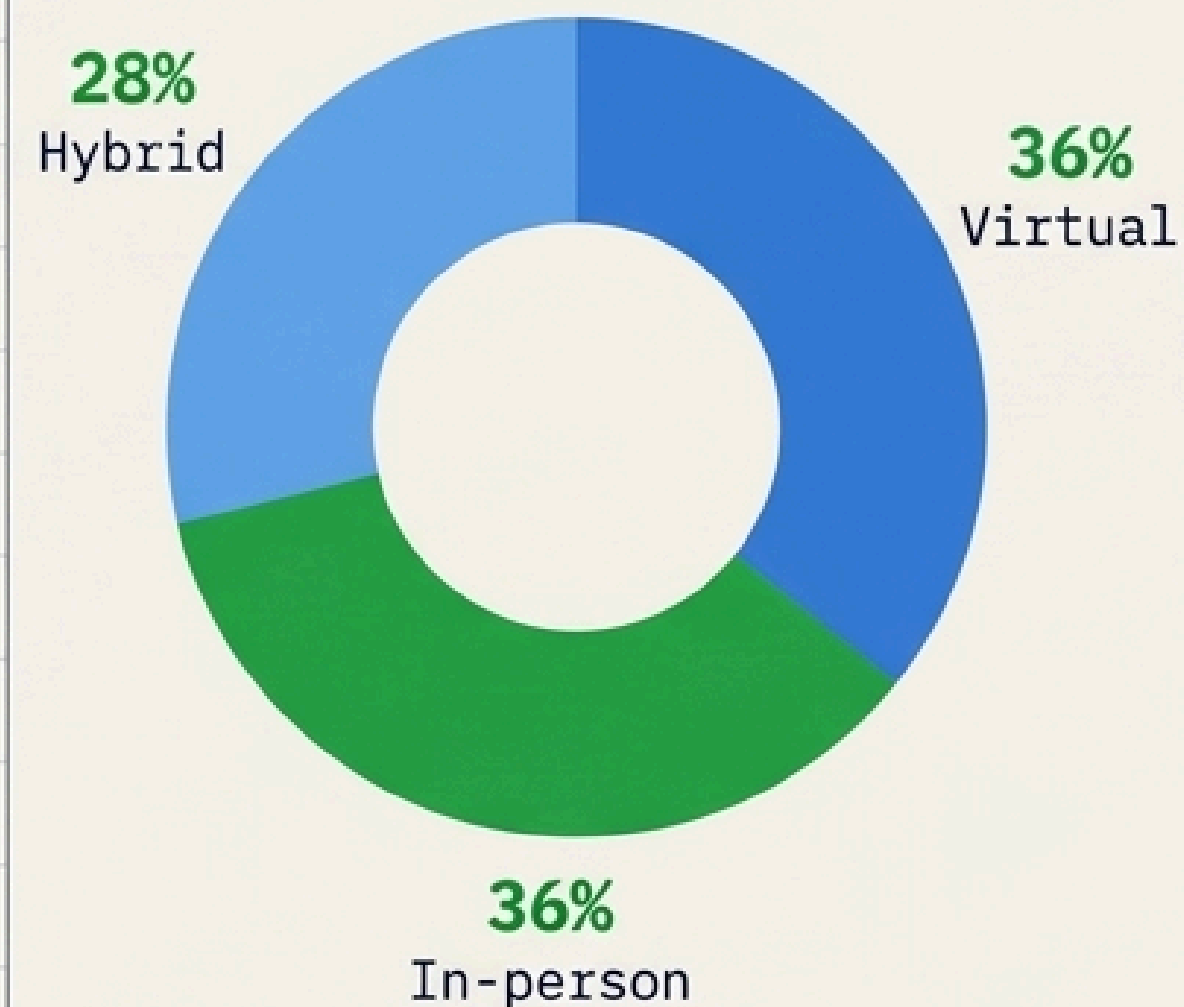
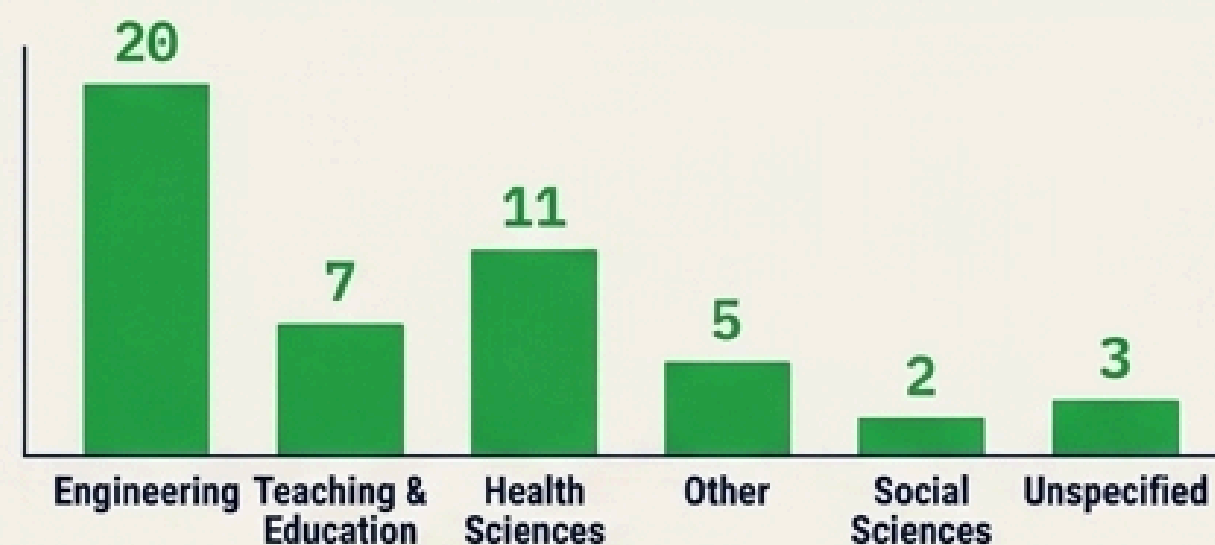
Constraint: Requires a hyper-efficient micro-scheduling engine.



40% Engineering/Math profiles



Constraint: High supply for technological and numerical projects.



Constraint: Simultaneous management of connectivity and geolocation metrics.

Arquitectura del Sistema: Coordinación de Voluntarios Tutores Basada en la Investigación

ENTRADAS DEL SISTEMA (Inputs)



Captura de Perfiles y Disponibilidad

Recopilación de datos de voluntarios, incluyendo área de estudio (40% ingeniería), competencias académicas y modalidad preferida (presencial/virtual).



Requerimientos de Proyectos

Identificación de las necesidades curriculares de las instituciones y los horarios específicos solicitados para el apoyo académico.

CAPA 1 - PRESENTACIÓN (Interfaces de Usuario)

Portal para Voluntarios



Interfaz donde los tutores gestionan su perfil, registran sesiones y consultan sus asignaciones.

Portal para Instituciones



Espacio para que las escuelas gestionen proyectos, horarios de colaboración y realicen seguimiento.

Seguimiento de Estudiantes



Módulo dedicado a visualizar el progreso académico y la experiencia de aprendizaje del beneficiario final.

CAPA 2 - APLICACIÓN (Lógica de Negocio)



Registro y Gestión

Módulo encargado de la incorporación formal de nuevos actores y la validación inicial de datos según la norma ISO 9001:2015.

COORDINACIÓN INTELIGENTE: ALGORITMO $C(v, p)$



$$C(v, p) = 0,40H_k + 0,30H_o + 0,15U_b + 0,15E_x$$

Núcleo del sistema que calcula la compatibilidad basada en: Habilidades (40%), Horario (30%), Ubicación (15%) y Experiencia (15%).



Seguimiento, Análisis y Notificaciones

Motor de mensajería automatizada y monitoreo constante del impacto académico para generar alertas tempranas de deserción o agotamiento.

CAPA 3 - DATOS (Persistencia)



API de Google Sheets

Repositorio temporal para la persistencia de información y memoria histórica del sistema.



Gestión de Riesgos (ISO 31000:2018)

Monitoreo de integridad de datos para mitigar riesgos de setoración de la API y asegurar la consistencia de los registros académicos.

FLUJO DE SALIDA (Outputs)



Asignaciones Optimizadas

Emparejamientos con puntajes de compatibilidad promedio entre 0,960 y 0,887, garantizando la mejor calidad educativa posible.

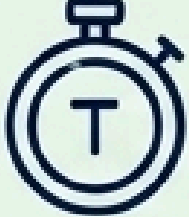
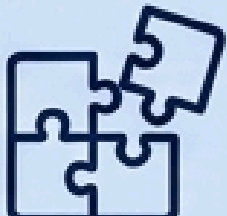
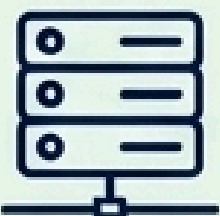


Reportes de Impacto y Alertas

Generación de métricas sobre cobertura educativa y estados de cuenta del voluntariado para la toma de decisiones.

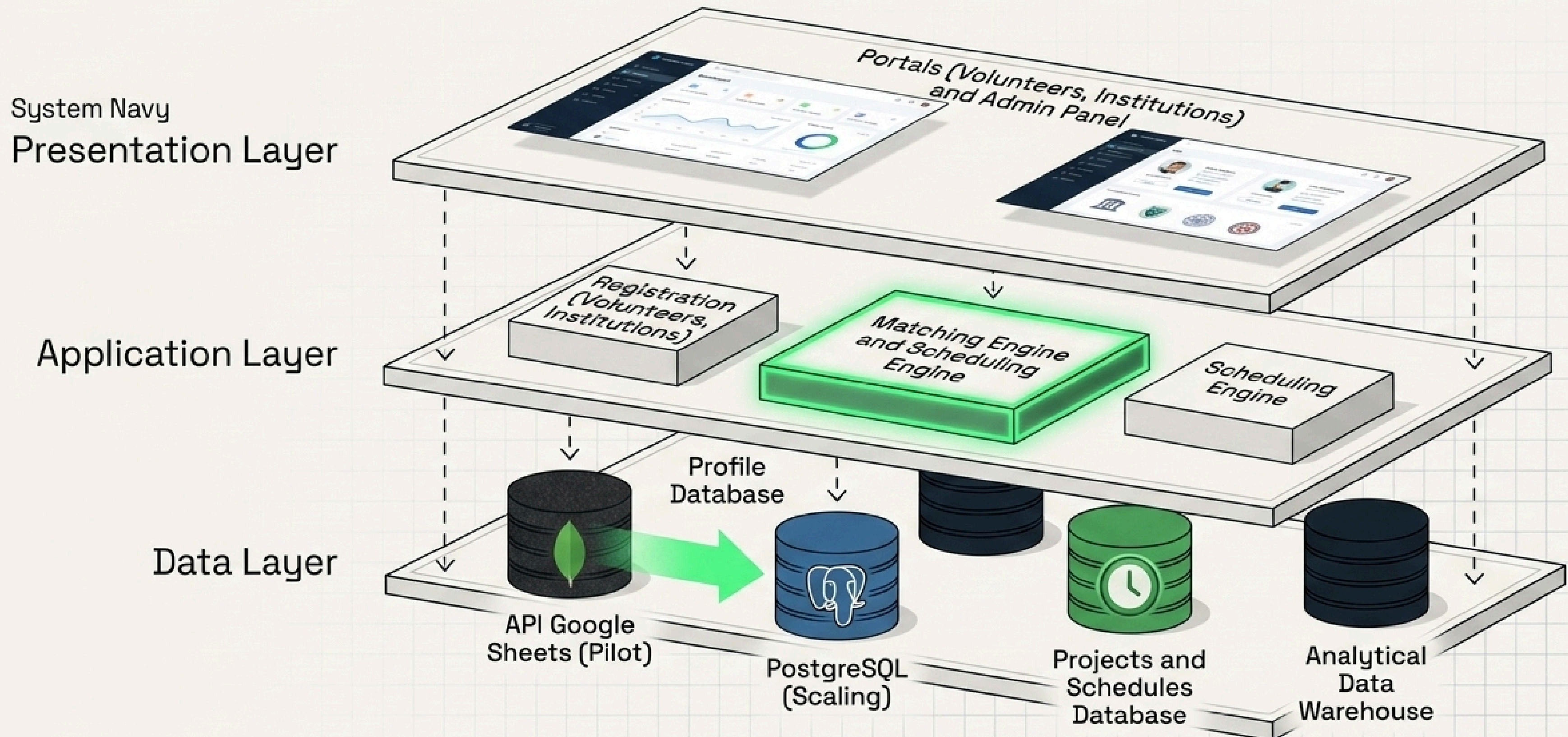
System Foundations: Critical Requirements (Workshop 2)

Defining success parameters for a scalable ecosystem.

Functional Requirements	Non-Functional Requirements
 Registration: Granular capture of profiles, competencies, and availability.	 Latency < 60s For the assignment engine.
 Matching: Mathematical algorithm for weighted compatibility.	 Availability > 99% Guaranteed monthly system uptime.
 Monitoring: Traceability of sessions and academic impact.	 80+ Users Guaranteed concurrent capacity.

Robust 3-Layer Architecture

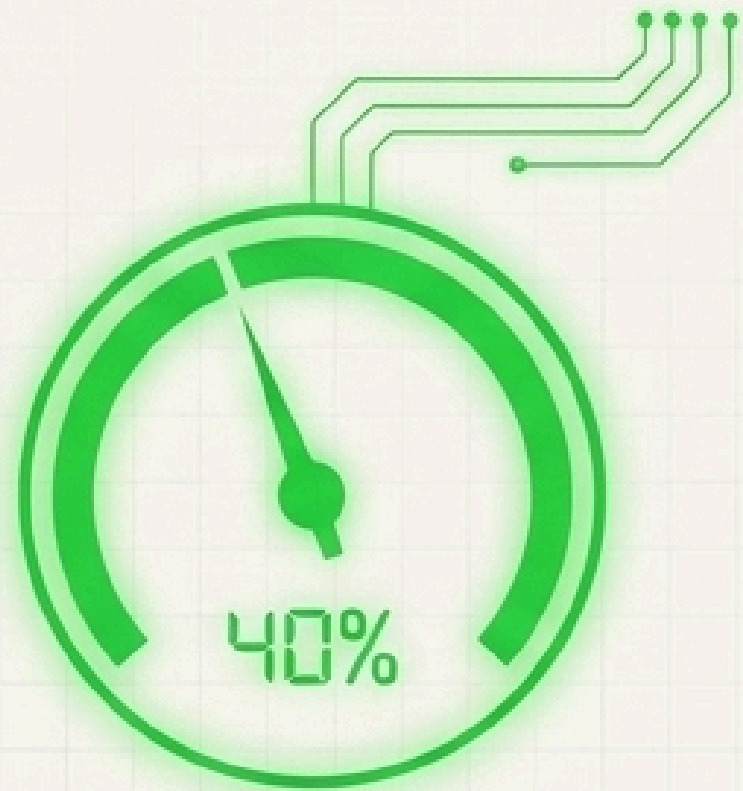
Modular design based on microservices and high fault tolerance.



The Core Engine: Weighted Compatibility Algorithm

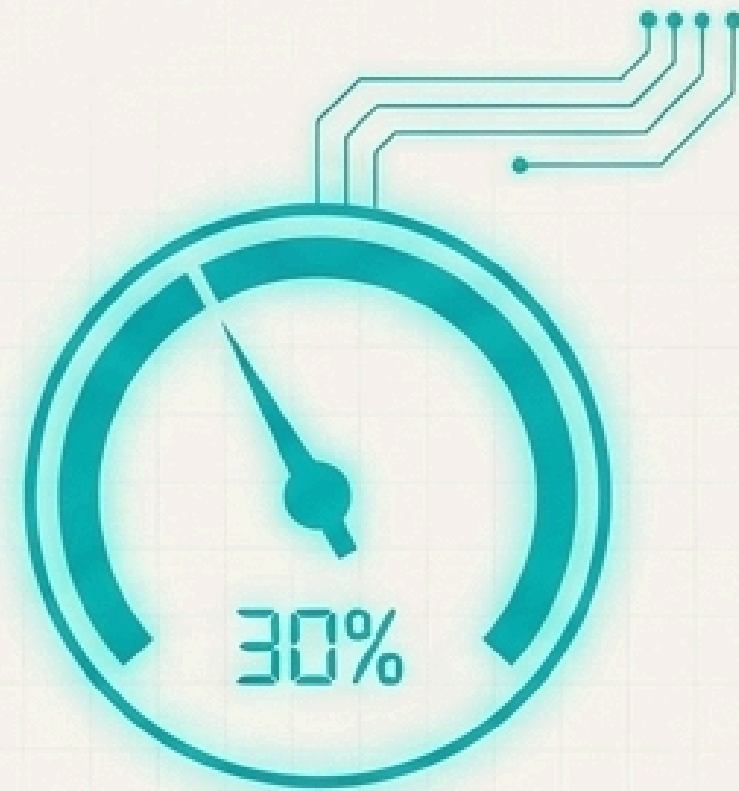
Mathematical quantification of human interaction.

$$C(v,p) = 0.40(Sk) + 0.30(Sc) + 0.15(SI) + 0.15(Se)$$



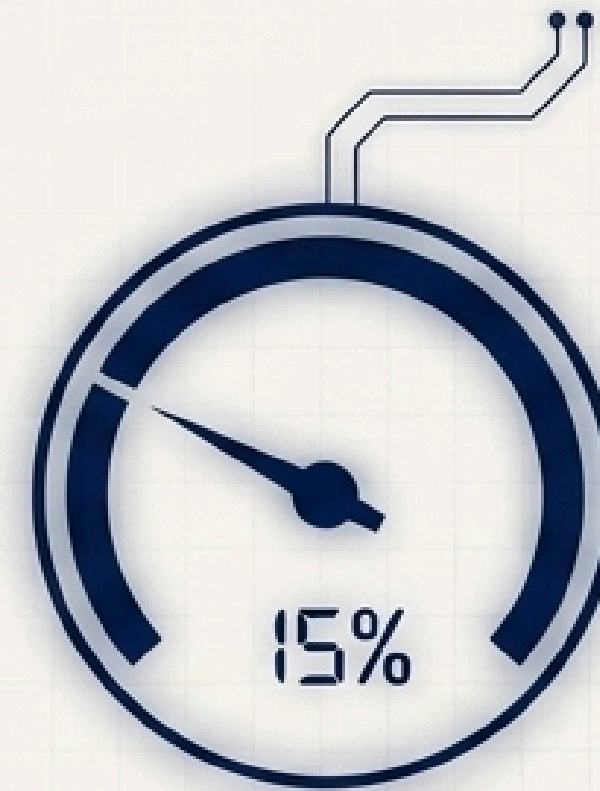
Academic Skill (Sk)

Dominant factor driven by tutoring needs.



Availability (Sc)

Mitigates the constraint of low volunteer time.



Proximity/Connectivity (SI)

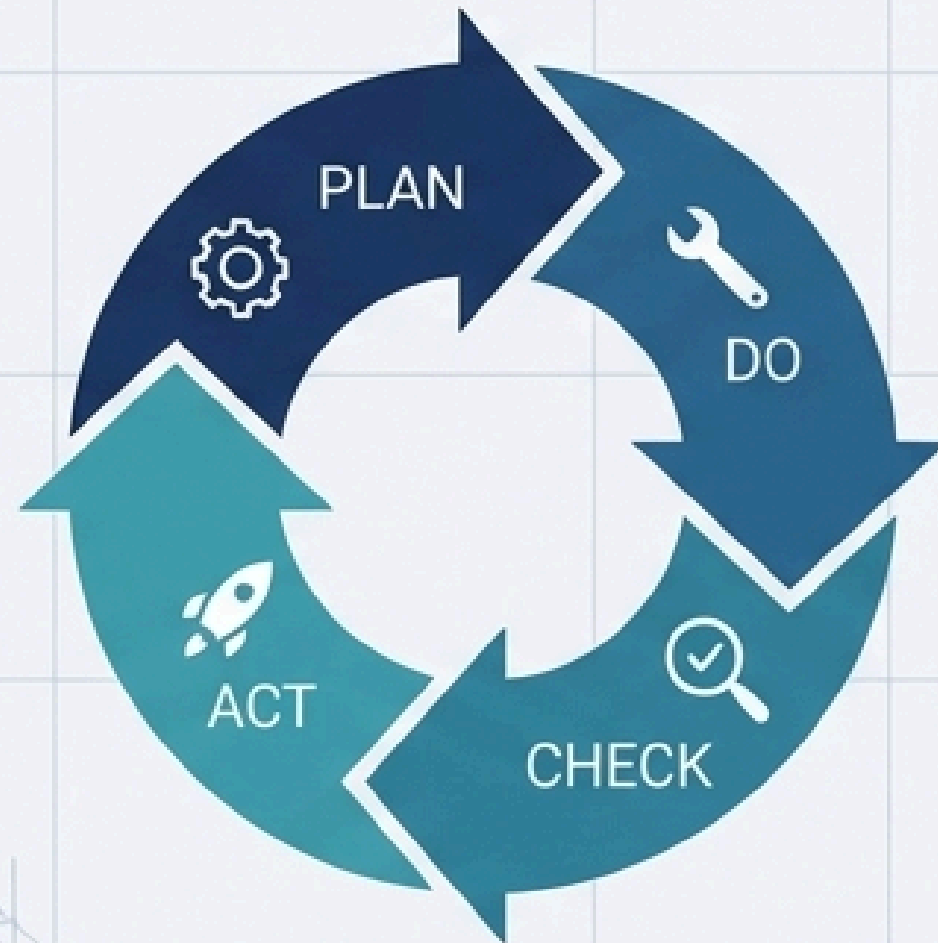


Pedagogical Experience (Se)

International Standards Framework (Workshop 3)

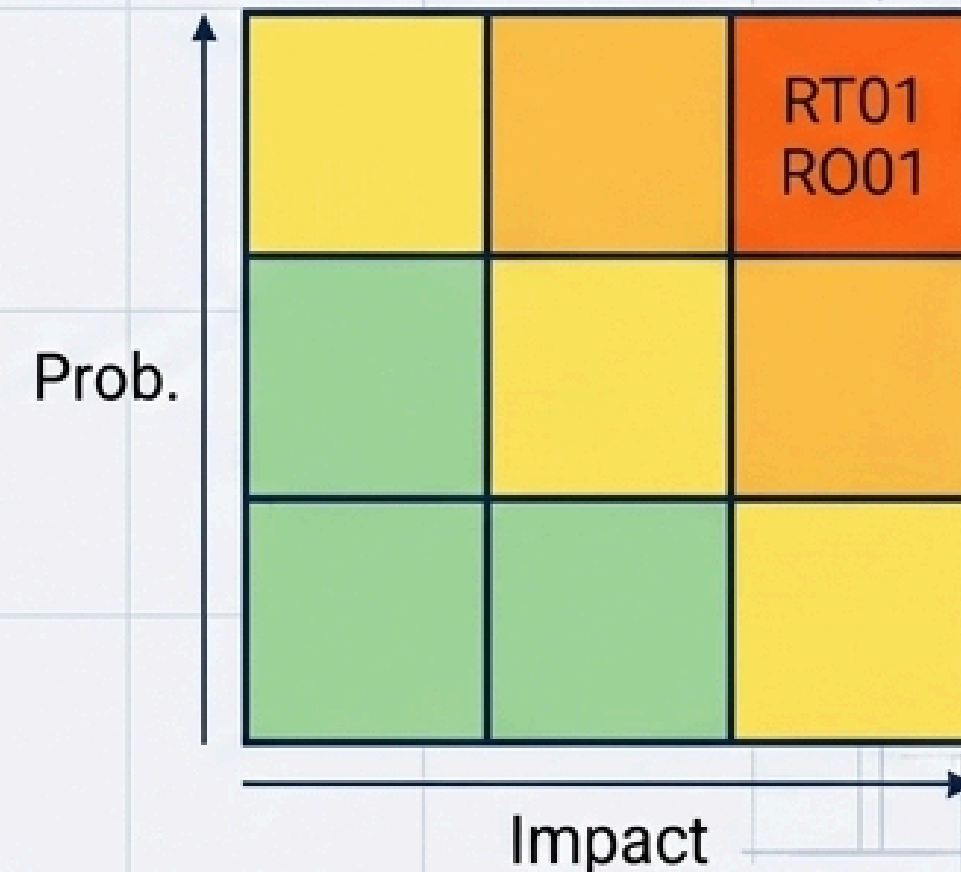
Quality assurance and proactive threat mitigation.

ISO 9001:2015 - Quality Management



PDCA cycle for continuous improvement and usability testing.

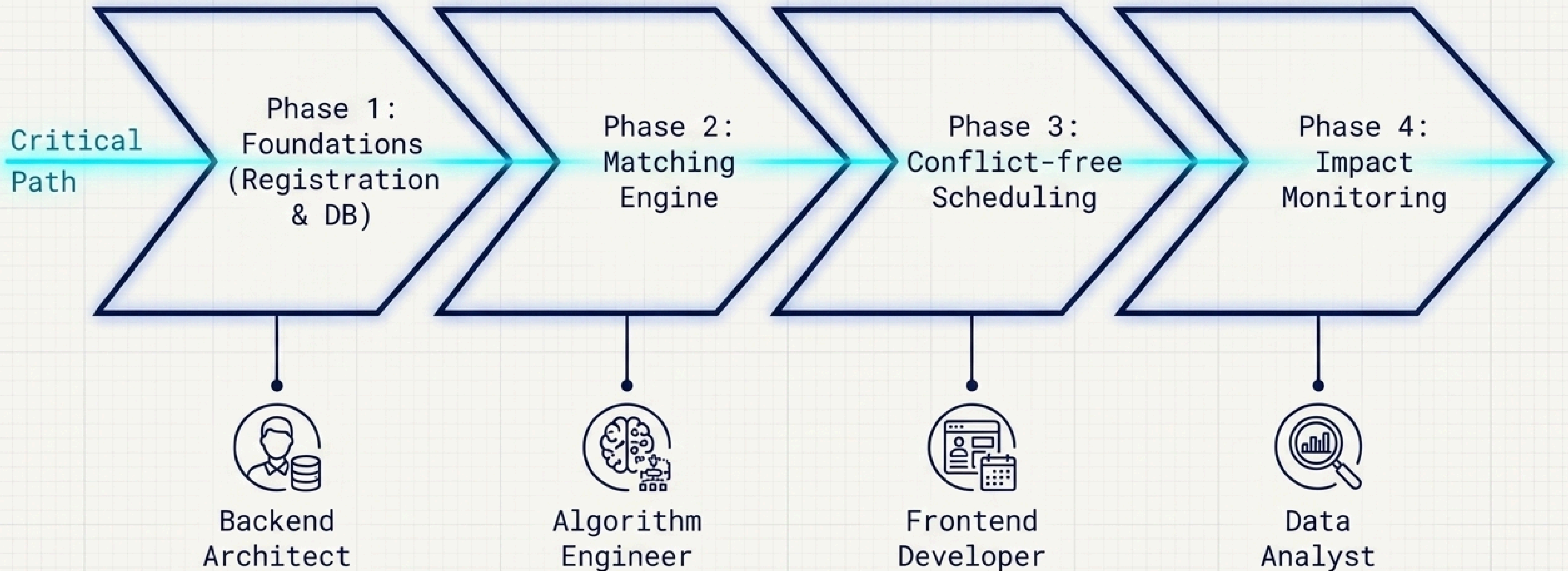
ISO 31000 - Risk Management



Heat matrix highlighting RT01 (Low data quality) and R001 (Volunteer shortage) as critical.

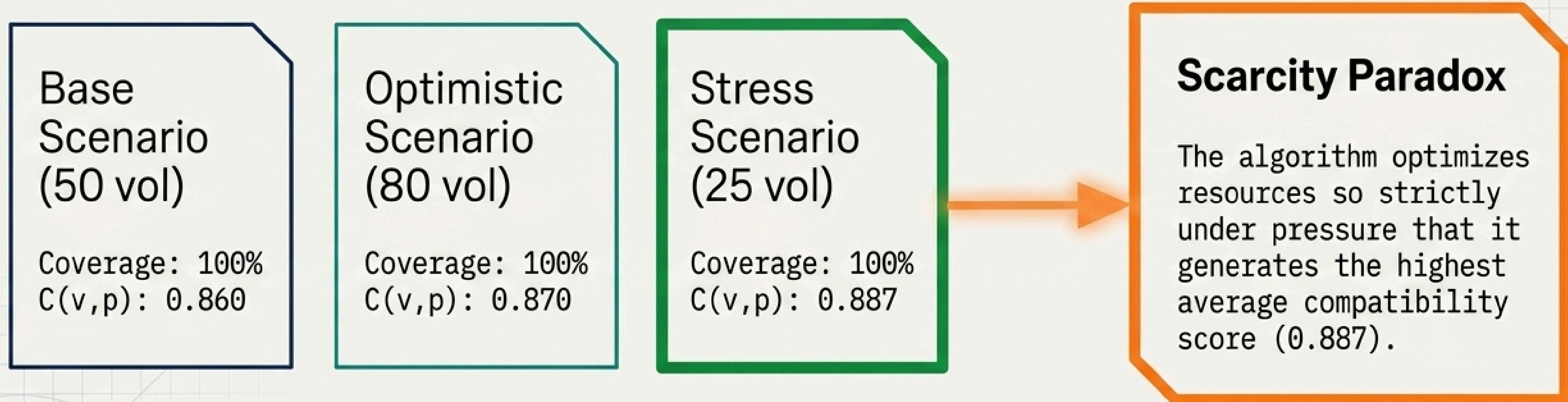
Execution Structure and Critical Path

Methodological traceability from design to deployment.



Simulation 1: Mechanical Validation (Discrete Events)

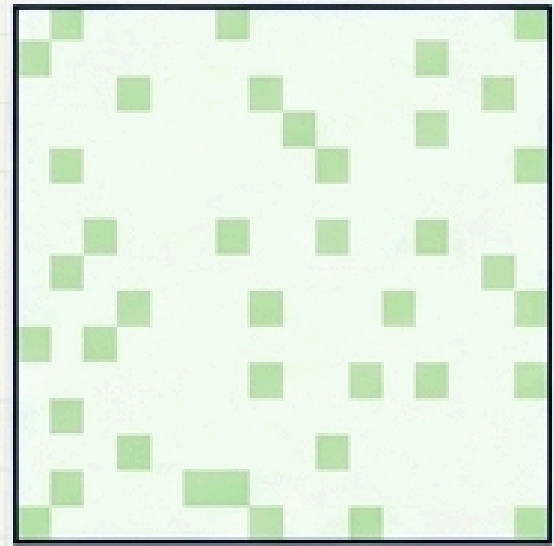
Modeling the assignment pipeline under operational stress (12 weeks).



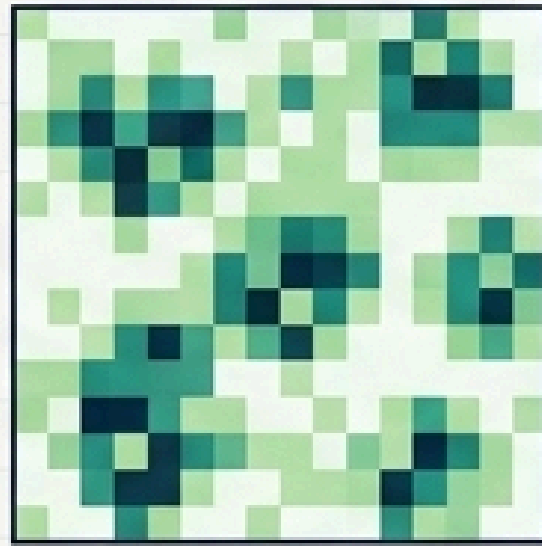
Simulation 2: Organic Validation (Cellular Automata)

The network amplification effect in complex adaptive ecosystems

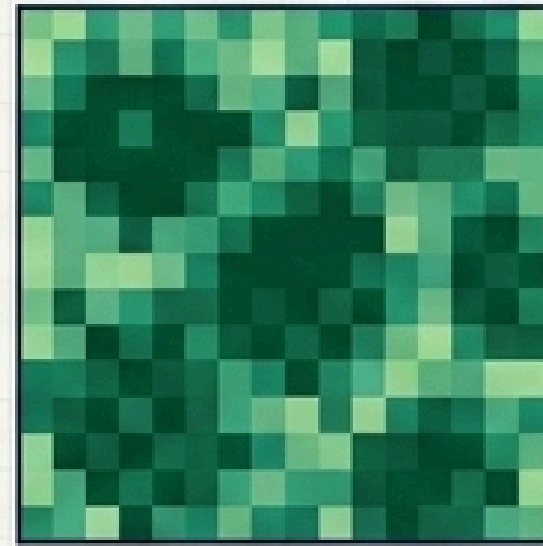
Active Platform (Weeks 0 to 20)



Week 0

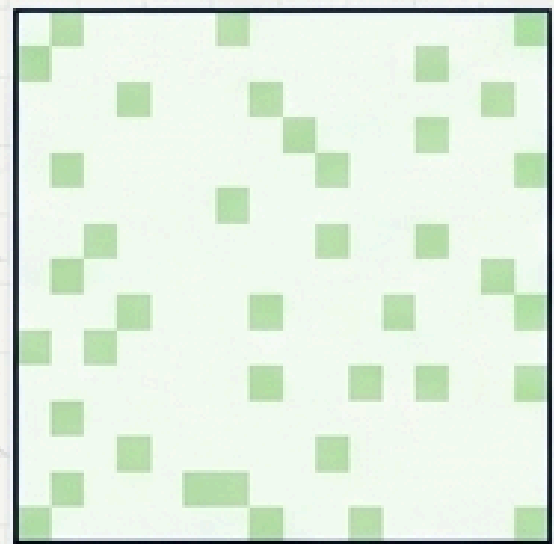


Week 10

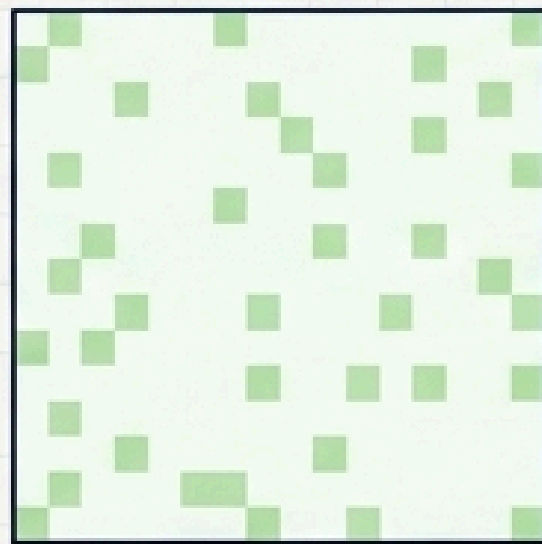


Week 20

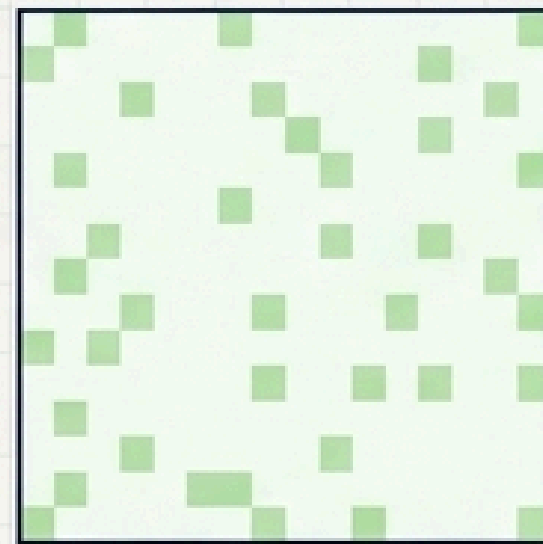
No Platform (Weeks 0 to 20)



Week 0



Week 10



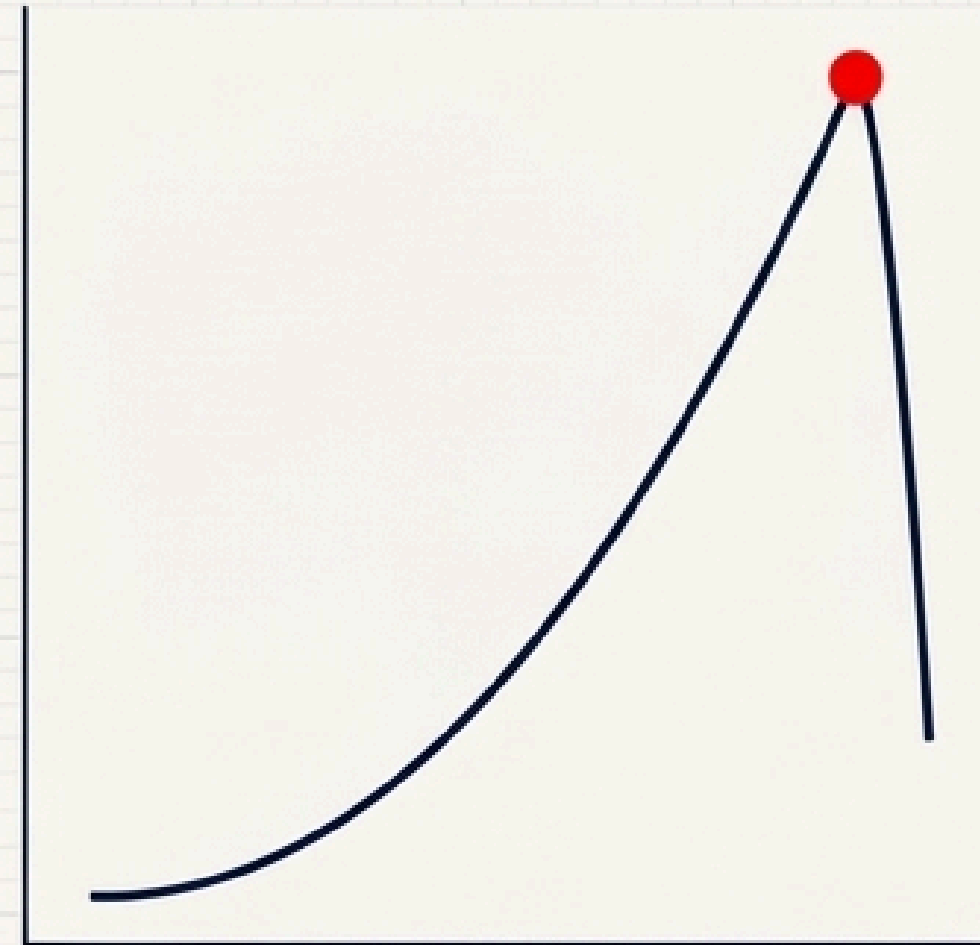
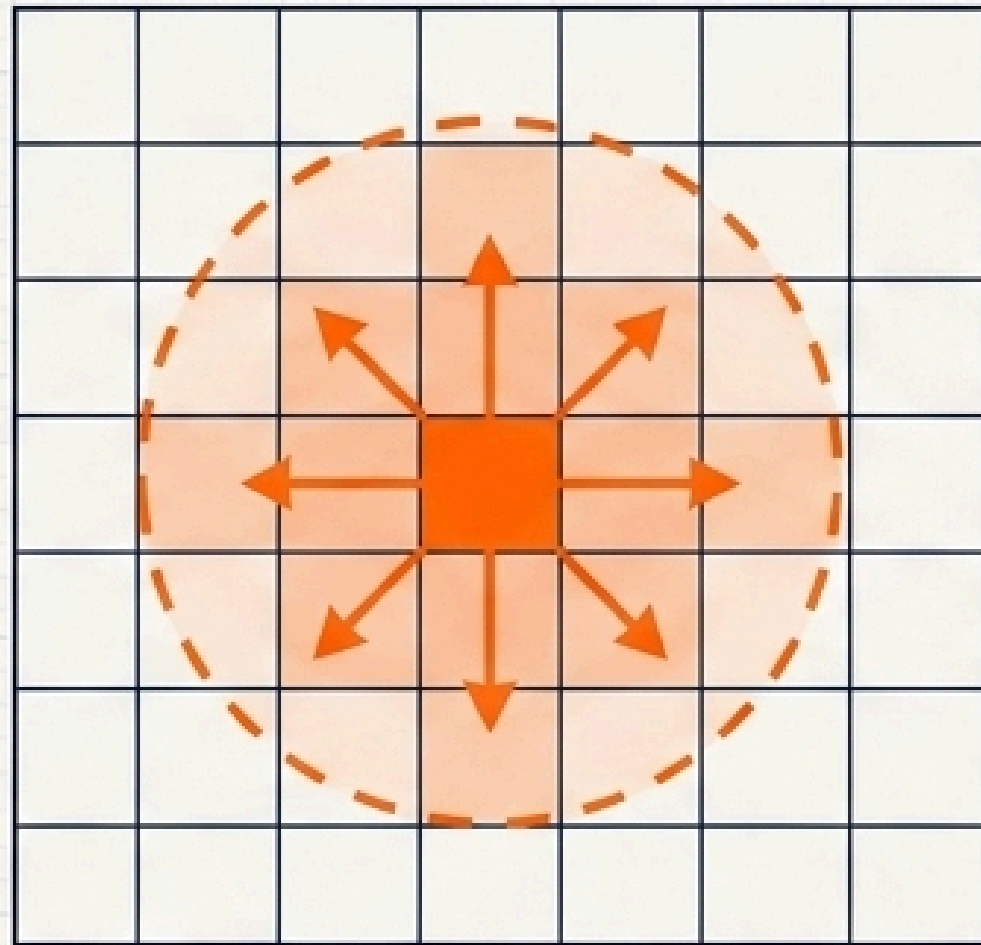
Week 20

+18%
Amplification

The platform acts as a social catalyst, mathematically accelerating engagement in the university community compared to a manual system.

Emergent Risk: The Burnout Contagion

Computational discovery of chaotic dynamics and the domino effect.



Critical Finding

Burnout spreads through social influence. An overloaded volunteer generates a cascade effect, putting the entire network's retention at risk.

Design Validation vs. Simulated Reality

Empirical verification of architectural requirements.

Design Goal (W2/W3)	Simulation Result (W4)	
Assignment Latency: < 60s	Average 45ms	
Project Coverage: > 70%	100% in all scenarios	
Compatibility Quality: > 0.80	Range 0.860 - 0.887	
Risk Discovery	Identification of Burnout (R005)	

Implementation Roadmap

Recommendations based on complex systems simulations.

1. Anti-Burnout Monitor

Implement automated alerts when exceeding 3 sessions/week.

2. Operation Baseline

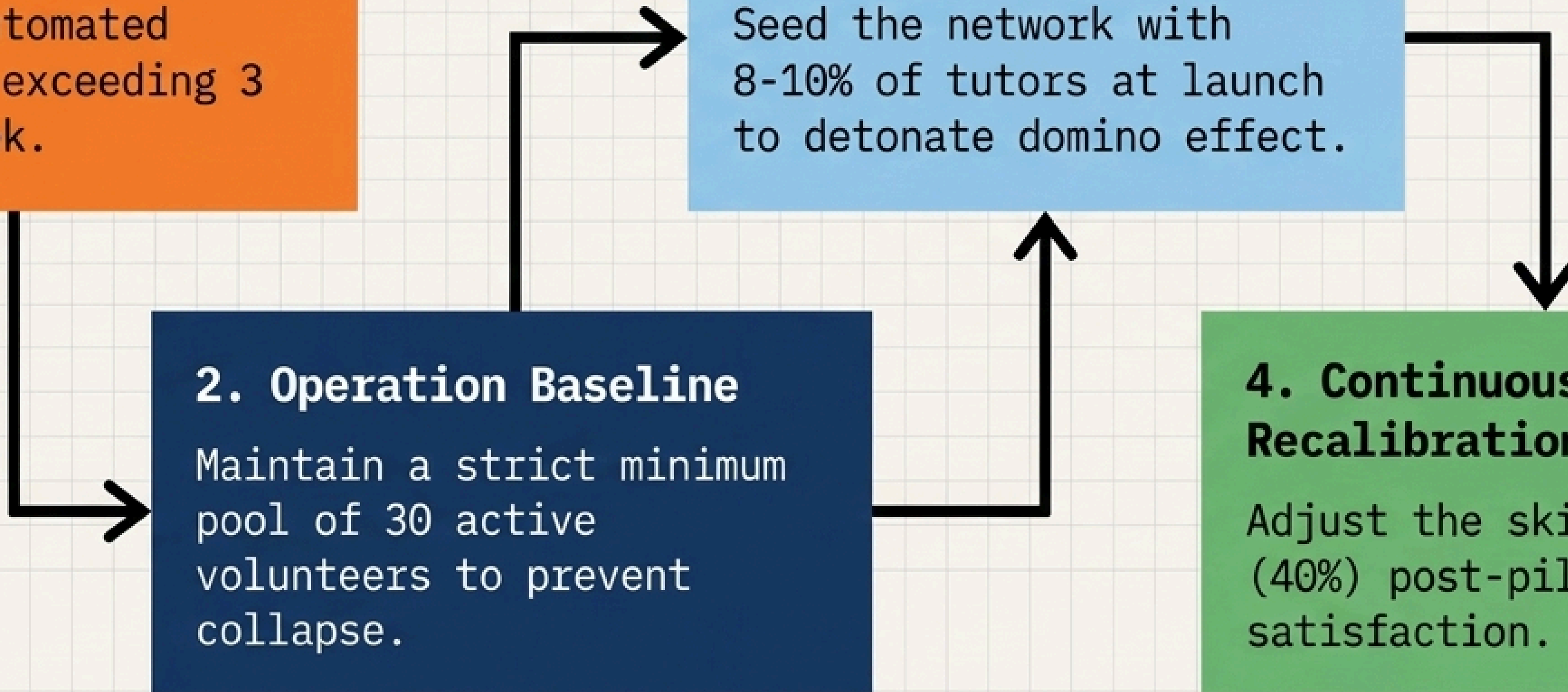
Maintain a strict minimum pool of 30 active volunteers to prevent collapse.

3. Critical Mass Strategy

Seed the network with 8-10% of tutors at launch to detonate domino effect.

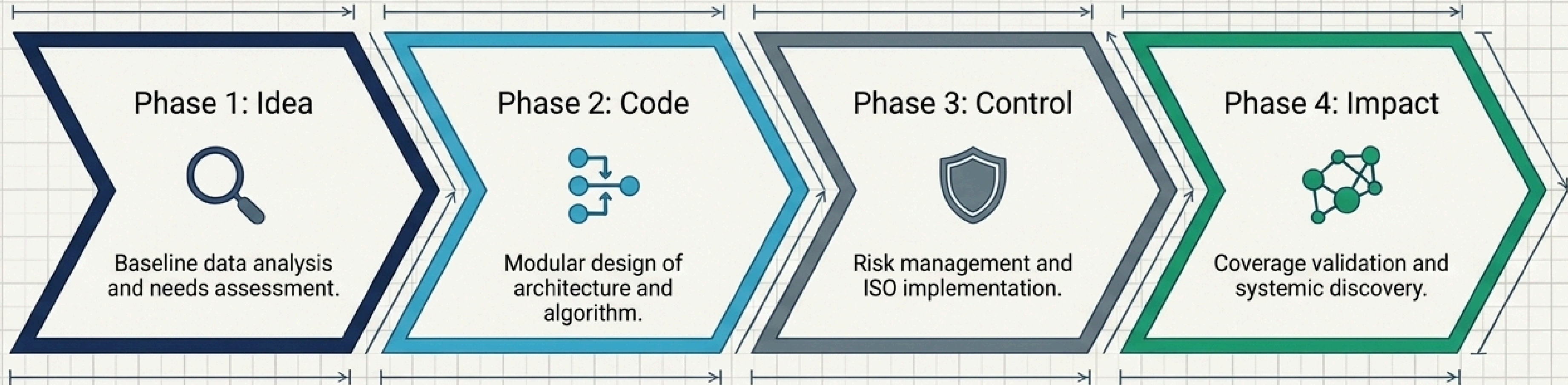
4. Continuous Recalibration

Adjust the skill weight (40%) post-pilot based on satisfaction.



“Engineering Lifecycle Conclusions” in System Navy

From a disorganized manual process to a mathematically validated digital ecosystem.



Educational Volunteer Coordination Platform

Systems engineering serving social impact.



Repository of Code, Simulations, and Documentation.